Challenge Brief Template

Digital Assistants: Evolution to 2030 (Public Sector)

1. Challenge summary

Challenge: Digital Assistants – Evolution to 2030 (Public Sector)

The "Digital Assistants: Evolution to 2030 (Public Sector)" challenge outlines a forward-looking vision for citizen-government interactions.

Participants are challenged to demonstrate how a next-generation digital assistant can reliably process unstructured or colloquial citizen inputs, translate complex administrative language into plain, easily accessible information, and seamlessly guide citizens through public service workflows.

2. Goal

Concept & Live Prototype (PoC) of a Digital Public Service Assistant in 2030

Objective: Development of a fully functional live demo showcasing the future of public sector interactions.

Key Focus Areas:

- *Overall Strategy & Customer Journey: End-to-end process design tailored for citizens.*
- *Design & Communication: Intuitive, clear, and accessible interfaces.*
- *Technical Architecture: A scalable and secure system foundation.*

Inclusion & Accessibility Commitment: Full accessibility for all demographic groups—seamlessly usable regardless of age, gender, digital literacy, or personal circumstances.

3. Team requirements

Only list what is truly required to solve the problem and say why. Delete this section if there are none.

- *Each team must include at least one developer (with focus/knowledge in artificial intelligence) and one UX/UI designer*
- *Knowledge in artificial intelligence as well as user experience design*

4. What Federal Chancellery provides

Be specific about access, limits and timing. Teams plan around this list.
Federal Chancellery will provide:

- Live-Demo (current solution and storyline – AI chatbot ida)
- Information regarding the target audience (citizens, public officials, etc.)

- Mentoring

Tech stack all hackathon's have access to (through sponsorships organised by the hackathon host):

- Complete tech stack from Cloudflare for all teams on enterprise level for one week.
- 10K credits from amazon web services

5. Participant responsibilities

List what teams must do, as actions. These become the basis for the deliverables and the evaluation.
Each team must (these are examples, include what is applicable)

- *Build a Prototype: Develop a functional live demo (or high-fidelity mockup).*
- *Design an End-to-End Customer Journey: Map and handle the complete user experience, with a clear focus on defined personas.*
- *Ensure High Usability: Maintain a user-centric design that keeps the experience as intuitive and accessible as possible.*
- *Operate Within Resource Constraints: Stay within the provided environment and budget (e.g., using the 10K in AWS credits—see Point 4).*

6. Expected behaviour

Show one realistic interaction or use case step by step.
The system should be able to handle scenarios such as:

- *please find some impressions at https://www.oesterreich.gv.at/en/formsearch and https://www.oesterreich.gv.at/en/landingpages*

***User / input:** Simulate a typical government interaction (e.g., passport renewal, address change, or permit application).*
***System:** the solution should support you within government services in an agent-based way*

The system must support:

- *German & English, focus on accessibility (all segments of the population)*
- *Fallback scenarios for digital immigrants (human contact point if desired)*

7. Suggested technical directions

Optional. Give a menu of approaches, not a prescribed solution. Teams stay free to choose.
Teams are free to choose their implementation approach. Possible techniques include:

- *Models & AI Frameworks: Open-source / open-weight LLMs, commercial APIs (e.g., OpenAI, Anthropic), or fine-tuned domain-specific models (if available).*

- *Open Data & APIs: Public registers, government open data portals, spatial data (e.g., OpenStreetMap), or synthetic citizen datasets.*

- *Inclusive Design & Accessibility:*
  Assistive tools (e.g., screen reader optimizations, plain-language generators, multi-language support) to ensure the solution serves all population segments.

8. Deliverables

Each team must deliver:

- ***High-Level Strategic Concept***
  - *An overarching strategy outlining the vision, target personas, and value proposition.*
- ***Working Prototype or High-Fidelity Mockup***
  - *A functional demonstration focused on 1–2 concrete use cases (e.g., automated forms, identity/passport updates, or public service guidance).*
  - *Inspiration & Reference Points: Explore existing public service portals at [oesterreich.gv.at Form Search] and [oesterreich.gv.at Topics & Services].*
- ***Short Technical Documentation A brief technical overview covering:***
  - *Architecture & Approach: System setup and key technical decisions.*
  - *Data & API Integration: How the required APIs and datasets are utilized.*
  - *Error Handling & Guardrails: How uncertainty, unexpected user inputs, or edge cases are managed.*
  - *Limitations & Edge Cases: Known constraints and potential areas for improvement.*
  - *Setup & Deployment: Step-by-step instructions to run or deploy the project.*
- ***Supporting Artifacts & Proof of Work** (where legally & technically feasible)*
  - *Code repository (e.g., GitHub / GitLab)*
  - *Short demo video (1–2 minutes) or interactive walkthrough*
  - *System logs and test results*

9. Evaluation

The evaluation will include:

- ***Pre-announced test cases:***
  - *Standard municipal workflows (e.g., renewing a passport, changing a primary residence, or inquiring about childcare subsidies).*
- ***Unannounced test cases:***
  - *Complex or combined requests to assess how effectively the model generalises across different administrative scenarios.*
- ***Realistic difficulties:***
  - *Handling noisy or ambiguous user input, colloquial language, incomplete citizen data, and edge cases (e.g., non-standard residency status).*
- ***Core tasks:***
  - *Accurately identifying intent, guiding citizens through multi-step forms, verifying prerequisites, and generating structured administrative output.*

The jury will assess whether the solution can [understand the input], [resolve the right result], [use your API / data] and [deliver a useful outcome].

Criterion	Weight	What the jury looks for
End-to-End Customer Journey & UX	[35%]	Delivers an intuitive, seamless flow from initial query to completed request; ensures full accessibility and inclusivity for all citizen demographics (age, language, digital literacy).
Innovation, Conversational Design & Accessibility	[25%]	Explores novel interaction patterns (e.g., voice/multimodal inputs, plain-language translation, proactive assistance) to make government services effortless, highly intuitive, and accessible for all citizens.
Core Accuracy & Solution Quality	[20%]	Correctly resolves citizen requests, accurately interprets complex intent, provides factually correct legal/administrative guidance, and delivers high-value outcomes without hallucinating.
Handling Uncertainty & Guardrails	[10%]	Gracefully detects missing information or ambiguity, asks clarifying questions, provides transparent confidence levels, and refrains from taking unauthorized administrative actions.
Robustness, Inclusivity & Polish	[10%]	Maintains stability under edge cases and noisy input; offers a polished UI/UX with smooth fallback mechanisms during API latency or system errors
Total	**100%**	

10. Constraints

- ***Simulated Environment:*** *Access to real government databases or live backend interfaces is not required. Solutions may use mock data, synthetic citizen profiles, or simulated APIs.*

- ***Public Sector Impact (Vision 2030):*** *The PoC must demonstrate how a 2030 digital*

assistant can bridge the needs of both citizens and public administration—streamlining procedures and reducing bureaucratic workload.

- ***Strategic Foundation:*** *Teams must submit a high-level strategy paper that outlines the overarching vision, target personas, and conceptual framework for the PoC.*

- ***Focus on Innovation:*** *The PoC should explicitly highlight innovative interaction patterns and novel approaches designed to radically simplify administrative workflows.*

11. Success definition

A successful project demonstrates that an inclusive digital assistant for Agenda 2030 can provide the public with reliable, understandable information (formulated in plain language) based on unstructured or conversational input, and utilize these results to facilitate seamless interactions within the public sector.

The solution aims to visualize future interactions between the public and the state and to demonstrate innovative new ways to alleviate the administrative burden.